Iowa State University

LAB 6 - MECHANICAL JOINTS REPORT

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Aer E 4260 Design of Aerospace Structures

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Contents

1

Introduction

In this lab, we focused on designing connections for various structural elements, emphasizing bolted and riveted joints subjected to shear and bearing stresses.

The first problem involved designing rivets for a gusset plate made of AISI 4340 steel. The challenge was to calculate the load distribution across the rivets and select appropriate rivet diameters to prevent failure under shear and bearing loads. Additionally, we verified the factor of safety for the rivet design to ensure it met the structural requirements.

In the second problem, we designed a shear clip to join an aluminum plate with a channel beam. The design required determining the optimal thickness of the clip and selecting rivets that could withstand the applied shear load. Proper spacing and configuration of the rivets were critical to ensure a secure connection between the two components.

The final problem involved evaluating the use of a pull-up fix for an I-beam with a vertical gap. We analyzed the forces acting on the rivets and calculated their safety margins to verify whether the pull-up solution would provide a reliable connection. The objective was to confirm the adequacy of the rivet selection and ensure the I-beam remained structurally sound.

2

Figures

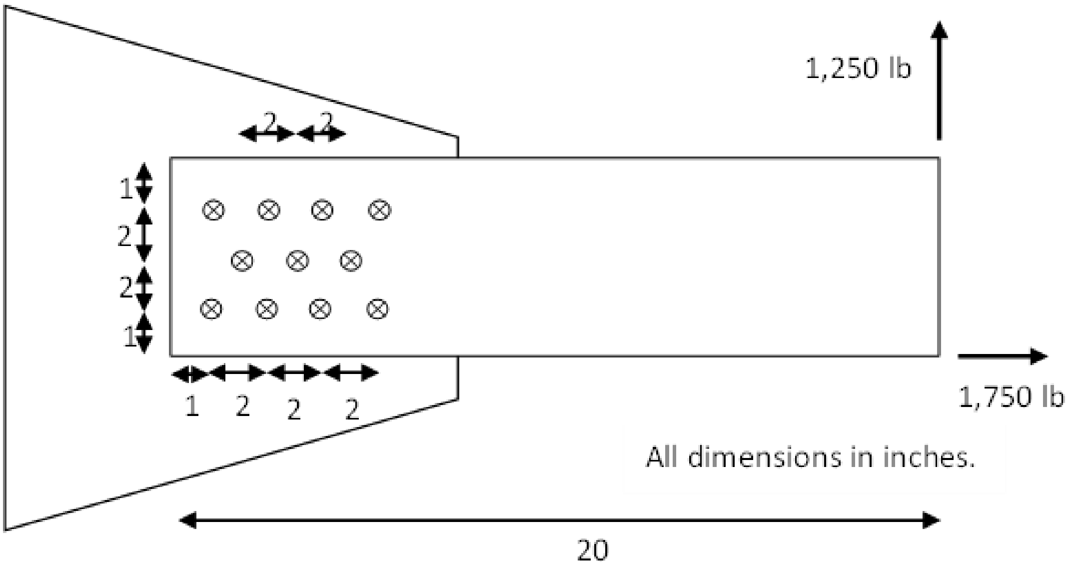


Figure 2.1: *Problem 1: Bolts in a gusset plate configuration and applied loads*

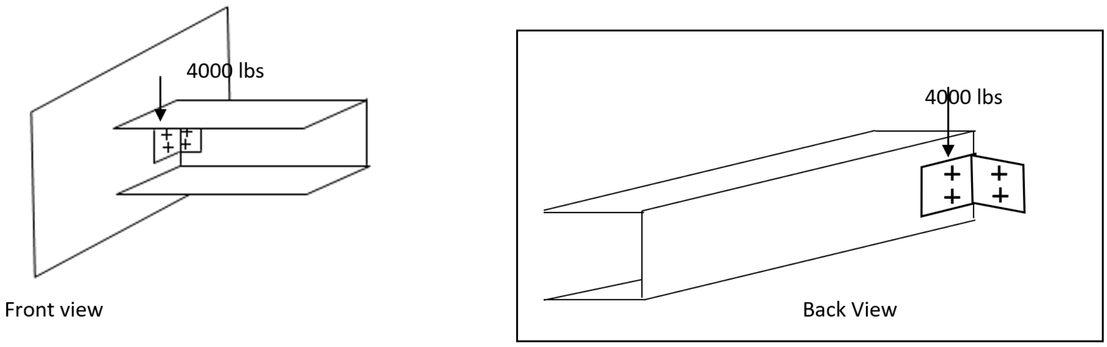


Figure 2.2: *Problem 2: Plate and channel beam connected by a shear clip front view (left) and back view (right)*

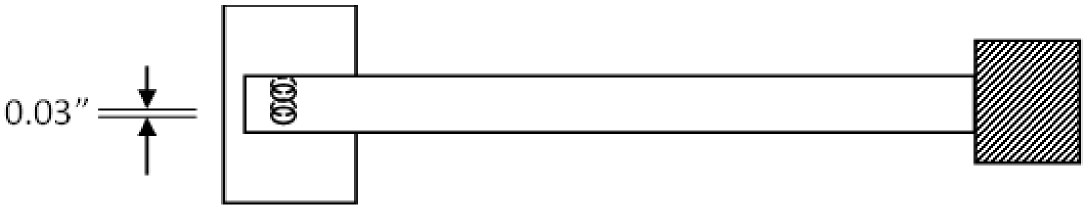


Figure 2.3: *Problem 3: I-Beam with vertical gap*

31 January 2003

Table 2.3.1.0(f_1). Design Mechanical and Physical Properties of Low-Alloy Steels

Alloy	Hy-Tuf	4330V	4335V	4335V	D6AC	AISI 4340[a]	0.40C 300M	0.42C 300M
Specification	AMS 6425	AMS 6411	AMS 6430	AMS 6429	AMS 6431	AMS 6414	AMS 6417	AMS 6257 AMS 6419
Form	Bar, forging, tubing							
Condition	Quenched and tempered[b]							
Thickness or diameter, in.	c				d	e	f	
Basis	S	S	S	S	S	S	S	S
Mechanical Properties:								
F_{tu}, ksi	220	220	205	240	220	260	270	280
F_{ty}, ksi	185	185	190	210	190	217	220	230
F_{cy}, ksi	193	193	199	220	198	235	236	247
F_{su}, ksi	132	132	123	144	132	156	162	168
F_{bru}, ksi								
(e/D = 1.5)	297	297	315	369	297	347	414[g]	430[g]
(e/D = 2.0)	385	385	389	465	385	440	506[g]	525[g]
F_{bry}, ksi								
(e/D = 1.5)	267	267	296	327	274	312	344[g]	360[g]
(e/D = 2.0)	294	294	327	361	302	346	379[g]	396[g]
e, percent:								
L	10	10	10	10	12	10	8	7
LT	5[a]	5[a]	7	7	9	...	...	...
E, 10^3 ksi	29.0							
E_c, 10^3 ksi	29.0							
G, 10^3 ksi	11.0							
μ	0.32							
Physical Properties:								
ω, lb/in.^{3}	0.283							
C, K, and α	See Figure 2.3.1.0							

a Applicable to consumable-electrode vacuum-melted material only.
b Design values are applicable only to parts for which the indicated F_{tu} has been substantiated by adequate quality control testing.
c Thickness ≤ 1.70 in. for quenching in molten salt at desired tempering temperature (martempering); ≤2.50 in. for quenching in oil at flow rate of 200 feet/min.
d Thickness ≤ 3.50 in. for quenching in molten salt at desired tempering temperature (martempering); ≤5.00 in. for quenching in oil at flow rate of 200 feet/min.
e Thickness ≤ 1.70 in. for quenching in molten salt at desired tempering temperature (martempering); ≤2.50 in. for quenching in oil at flow rate of 200 feet/min.; ≤3.50 in. for quenching in water at a flow rate of 200 feet/min.
f Thickness ≤5.00 in. for quenching in oil at a flow rate of 200 feet/min.
g Bearing values are "dry pin" values per Section 1.4.7.1.

Table 2.1: *Design Mechanical and Physical Properties of Low-Alloy Steels*

Table 8.1.2(b). Single Shear Strength of Solid Rivets[a]

Undriven			Driven			Rivet Size							
Rivet Material	F_{su} (ksi) Min	F_{su} (ksi) Max	Rivet Material	F_{su}^{c} (ksi)	Rivet Designation	1/16	3/32	1/8	5/32	3/16	1/4	5/16	3/8
						Driven Single Shear Strength, lbs[b]							
5056-H32	24	n/a	5056-H321[d]	28[e]	B[f]	99	203	363	556	802	1450	2290	3275
2117-T4	26	n/a	2117-T3	30[e]	AD	106	217	389	596	860	1555	2455	3510
2017-T4	35	42	2017-T3	38[e]	D	134	275	493	755	1085	1970	3115	4445
2024-T4	37	n/a	2024-T31	41[g]	DD	145	297	532	814	1175	2125	3360	4795
7050-T73	41	46	7050-T731[d]	43[e]	E[h]	152	311	558	854	1230	2230	3520	5030
Monel	49	59	Monel	52[e]	M	183	376	674	1030	1490	2695	4260	6085
Ti-45Cb	50	59	Ti-45Cb	53[e]	T	187	384	687	1050	1515	2745	4340	6200
A-286	85	95	A-286	90[e]	-	317	651	1165	1785	2575	4665	7375	10500

a All rivets must be sufficiently driven to fill the rivet hole at the shear plane. Driving changes the rivet strength from the undriven to the driven condition and thus provides the above driven shear strengths.
b Based on nominal hole diameter specified in Table 8.1.2(a).
c Shear stresses are for the as driven condition on B-basis probability.
d The temper designations last digit (1), indicates recognition of strengthening derived from driving.
e The bucktail's minimum diameter is 1.5 times the nominal hole diameter in Table 8.1.2(a).
f Should not be exposed to temperatures over 150°F.
g Driven in the W (fresh or ice box) condition to minimum 1.4D bucktail diameter.
h E (or KE, as per NAS documents).

Table 2.2: *Single Shear Strength of Solid Rivets*

Table 8.1.2.1(a). Unit Bearing Strength of Sheet on Rivets, F_{br} = 100 ksi

Sheet thickness, in.	Unit Bearing Strength for Indicated Rivet Diameter, lbs 1/16	3/32	1/8	5/32	3/16	1/4	5/16	3/8
0.012	80	...	...	...	...	...	...	...
0.016	107	...	...	...	...	...	...	...
0.018	121	173	...	...	...	...	...	...
0.020	134	192	...	...	...	...	...	...
0.025	168	240	321	...	...	...	...	...
0.032	214	307	411	509	...	...	...	...
0.036	241	346	462	572	688	...	...	...
0.040	268	384	514	636	764	...	...	...
0.045	302	432	578	716	860	...	...	...
0.050	335	480	642	795	955	1285	...	...
0.063	422	605	810	1002	1203	1619	2035	...
0.071	476	682	912	1129	1356	1825	2293	2741
0.080	536	768	1028	1272	1528	2056	2584	3088
0.090	603	864	1156	1431	1719	2313	2907	3474
0.100	670	960	1285	1590	1910	2570	3230	3860
0.125	838	1200	1606	1988	2388	3212	4038	4825
0.160	1072	1536	2056	2544	3056	4112	5168	6176
0.190	1273	1824	2442	3021	3629	4883	6137	7334
0.250	1670	2400	3210	3975	4775	6425	8075	9650

Table 2.3: *Unit Bearing Strength of Sheet on Rivet*

3 Methodology

3.1 Problem 1

For problem 1, we were prompted to design the bolts in a gusset plate made of AISI 4340 steel, with a thickness of 1/4 in. The beam being connected is 6 in wide and 3/8 in thick. The loads and rivets positions can be observed in Figure 2.1. To solve this problem, a MATLAB code was developed (Appendix B).

The first step was to calculate the loads on each rivet, multiplying the applied loads by the factor of safety of 5. With MATLAB, I first calculated $\bar{X} = 4$ in and $\bar{Y} = 3$ in, taking into consideration that the origin (0,0) is at the bottom-left corner of the plate. I've also initially assume a value for the rivets diameter, $D = 3/8$ in.

With the known values for $\bar{X}$, $\bar{Y}$, and D, the moment of inertia was calculated $I = 30\ \text{in}^4$, and consequently the loads in each rivet. The maximum load on all rivets was found to be $P_{\text{max}} = 6613.0989$ lbs.

Now, going to Table 2.2, based on P_{max}, the rivet with 5/16 in diameter and made of A-286 was chosen ($F_{rivet} = 7375$ lbs). With the actual maximum load at a single rivet, the FS with respect to the rivet could be calculated:

$$P_{\text{max, actual}} = \frac{P_{\text{max}}}{\text{FS}} = \frac{6613.0989 \text{ psi}}{5} = 1322.6 \text{ lbs} \tag{3.1}$$

$$\text{FS}_{\text{rivet}} = \frac{F_{rivet}}{P_{\text{max, actual}}} = \frac{7375 \text{ lbs}}{1322.6 \text{ lbs}} = 5.58 \tag{3.2}$$

To check for bearing failure, first we need to calculate e/D:

$$e/D = \frac{1 \text{ in}}{5/16 \text{ in}} = 3.2 \tag{3.3}$$

With e/D, we can retrieve the yield bearing stress from Table 2.1: $F_{\text{br,y}} = 346$ ksi. Lastly, we can conver to an allowable bearing load and calculate the bearing factor of safety:

$$P_{\text{allow}} = F_{\text{br,y}} \cdot t = (346000 \text{ psi}) \cdot 3/8 \text{ in} = 40547 \text{ lbs} \tag{3.4}$$

$$\text{FS}_{\text{br}} = \frac{P_{\text{allow}}}{P_{\text{max, actual}}} = \frac{40547 \text{ lbs}}{1322.6 \text{ lbs}} = 30.7 \tag{3.5}$$

Thus, the plate won't fail in bearing.

3.2 Problem 2

In problem two, we are prompted to design a shear clip to join a plate with a channel beam at a right angle. Thus, our objective is to design the thickness of the aluminum sheet that will be used for the clip and the four rivets (two on the plate and two on the beam). From the problem statement, we know that the total shear load is $P = 4000$ lbs, and the aluminum channel high is $h_{\text{channel}} = 4$ in. Based on that height, I decided on the dimensions of the clip to be $h_{\text{clip}} = 3$ in (height of clip), $d_v = 1$ in (distance in between rivets), and $d_h = 0.5$ in (distance from right angle to rivet). Thus, I could calculate R and consequently F.

$$\sum M_0 = P \cdot d_h = R \cdot d_v \quad \Rightarrow \quad R = \frac{(4000 \text{ lbs}) \cdot (0.5 \text{ in})}{(1 \text{ in})} = 2000 \text{ lbs} \tag{3.6}$$

$$F = \sqrt{\left(\frac{(4000 \text{ lbs})}{2}\right)^2 + (2000 \text{ lbs})^2} = 2828.43 \text{ lbs} \tag{3.7}$$

Knowing the load F coming on each rivet, we could choose the solid rivet from Table 2.2. The chosen rivet is 5/16 in size and made of 2017-T4 material. Its shear strength is $\tau = 3115$ lbs. For the thickness of the sheet, considering F and the chosen rivet diameter, $t = 0.090$ in was chosen from Table 2.3, with a bearing strength of 2907 lbs. However, the table displays data for sheets with a bearing stress of 100 ksi, and our design has a bearing stress of 294 ksi. Thus, we need to adjust our thickness relative to the ratio of the bearing stresses:

$$t = \frac{0.090 \text{ in}}{2.94} = 0.0306 \text{ in} \tag{3.8}$$

We can choose the closest thickness available from Table 2.3, $t = 0.032$ in. Lastly, with our rivet and thickness defined, we can calculate the factors of safety:

$$\text{FS}_{\text{rivet}} = \frac{\tau_{\text{rivet}}}{F} = \frac{3115 \text{ lbs}}{2828.43 \text{ lbs}} = 1.10 \tag{3.9}$$

$$\text{FS}_t = \frac{\tau_t}{F} = \frac{2907 \text{ lbs}}{2828.43 \text{ lbs}} = 1.03 \tag{3.10}$$

3.3 Problem 3

For problem 3, we need to figure out if it is safe to use a pull-up with FS of 1.5 to fiz an I-beam with a vertical gap of 0.03 in, and to design the piece. The beam is 18 in long, with a depth $d = 3$ in, web thickness $t_w = 0.17$ in, flange width $b = 2.33$ in, flange thickness $t_f = 0.17$ in, cross-section area $A = 1.3$ in^2, and moment of inertia $I = 1.85$ in^4. Its material is 4330V low-alloy steel, with $E = 2.9 \times 10^3$ ksi retrieved from Table 2.1. Initially, we can calculate the residual tension stress on the beam F_t and the vertical load P at the end with a gap:

$$F_t = \frac{3.0 \cdot (29 \times 10^3 \text{ ksi}) \cdot (3 \text{ in}) \cdot (0.03 \text{ in})}{2 \cdot (18 \text{ in})^2} = 12.083 \text{ ksi} \tag{3.11}$$

$$P_{\text{max}} = \text{FS} \cdot \frac{3EIy_{\text{max}}}{L^3} = 1.5 \cdot \frac{3 \cdot (29 \times 10^3 \text{ ksi}) \cdot (1.85 \text{ in}^4) \cdot (0.03 \text{ in})}{(18 \text{ in})^3} = 1.242 \text{ kip} \tag{3.12}$$

With P_{max}, we can calculate the moment M at the end with a gap, and consequently R, knowing the distance in between rivets $d_v = 2$ in from the problem statement:

$$M = \frac{(1.242 \text{ kip}) \cdot (18 \text{ in})}{2} = 11.177 \text{ kip} \cdot \text{in} \tag{3.13}$$

$$M = R \cdot d_v \quad \Rightarrow \quad R = \frac{(11.177 \text{ kip} \cdot \text{in})}{(2 \text{ in})} = 5.589 \text{ kip} \tag{3.14}$$

Now with P_{max} and R, we can calculate F:

$$F = \sqrt{\left(\frac{(1.242 \text{ kip})}{2}\right)^2 + (5.589 \text{ kip})^2} = 5.623 \text{ kip} \tag{3.15}$$

Knowing F, the rivet with size 3/8 in, Monel material, and shear strength $\tau = 6085$ lbs was chosen from Table 2.2.

To find the bearing stress for the piece material, we need to calculate e and consequently e/D. The height of the piece was chosen to be $h = 4$ in.

$$e = \frac{h - d_v}{2} = \frac{4 \text{ in} - 2 \text{ in}}{2} = 1 \text{ in} \tag{3.16}$$

$$e/D = \frac{1 \text{ in}}{3/8 \text{ in}} = 2.67 \tag{3.17}$$

From Table 2.1, for the 4330V low-alloy steel with $e/D = 2.0$ (closest to 2.67), we get $F_{\text{br,y}} = 294$ ksi. Now, knowing F and the rivet's diameter, $t = 0.160$ in was chosen from Table 2.3, with a bearing strength of 6176 lbs. Applying the adjustment relative to the actual bearing stress of 294 ksi:

$$t = \frac{0.160 \text{ in}}{2.94} = 0.054 \text{ in} \tag{3.18}$$

We can choose the closest thickness available from Table 2.3, $t = 0.063$ in. With our rivet and thickness defined, we can calculate the actual F and the factors of safety:

$$F_{\text{actual}} = \frac{F}{\text{FS}} = \frac{5.623 \text{ kip}}{1.5} = 3.749 \text{ kip} \tag{3.19}$$

$$\mathrm{FS}_{\mathrm{rivet}} = \frac{\tau_{\mathrm{rivet}}}{F_{\mathrm{actual}}} = \frac{6.085\ \mathrm{kip}}{3.749\ \mathrm{kip}} = 1.62 \tag{3.20}$$

$$\mathrm{FS}_t = \frac{\tau_t}{F_{\mathrm{actual}}} = \frac{6.176\ \mathrm{kip}}{3.749\ \mathrm{kip}} = 1.65 \tag{3.21}$$

Lastly, we can compare the pullup stress F_t against the beam's material strength (from Table 2.1) to ensure the displacement doesn't induce failure in the beam itself.

$$\mathrm{FS}_{\mathrm{pullup}} = \frac{F_{t,y}}{F_t} = \frac{185\ \mathrm{ksi}}{12.083\ \mathrm{ksi}} = 15.3 \tag{3.22}$$

Concluding, it would be safe to use the pullup.

4

Conclusion

In this report, we successfully designed and evaluated the connections for three structural elements. For the gusset plate in Problem 1, rivets with appropriate diameters were selected to ensure both shear and bearing safety factors exceeded the required limits. In Problem 2, the shear clip and rivet design met the necessary load requirements with acceptable factors of safety. Finally, in Problem 3, it was confirmed that a pull-up fix could safely address the vertical gap in the I-beam, and the rivet configuration was validated to support the applied loads.

A

Appendix A

A.1 Rough Work

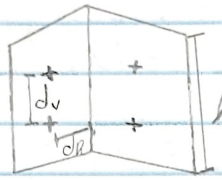
AERE 426 - Lab Assignment 6

Problem 2) t_{clip}, d_{rivets}? $h_{channel} = 4\ in$ $P = 4000\ lbs$

$h_{clip} = 3\ in$

$d_v = 1\ in$

$d_h = 0.5\ in$



$\Sigma M_o = P \cdot d_h = R \cdot d_h =$

$R = \dfrac{(4000\ lbs)(0.5\ in)}{(1\ in)}$

$R = 2000\ lbs$

$F = \sqrt{\left(\dfrac{P}{2}\right)^2 + R^2} = \sqrt{\left(\dfrac{(4000\ lbs)}{2}\right)^2 + (2000\ lbs)^2} = 2828.43\ lbs$

$\therefore$ 5/16 rivet chosen

$R_{F_{br}} = \dfrac{294\ ksi}{100\ ksi} = 2.94$ $t_{ess} = \dfrac{0.09\ in}{2.94} = 0.032\ in$

$FS_{rivet} = \dfrac{\tau_{rivet}}{F} = \dfrac{3115\ lbs}{2828.43\ lbs} = 1.10$ //

$FS_t = \dfrac{\tau_t}{F} = \dfrac{2907\ lbs}{2828.43} = 1.03$ //

Problem 3) $y = 0.03\ in$ $E = 29 \times 10^3\ ksi$ $L = 18\ in$

$d = 3\ in$

$F_t = \dfrac{3(29\times10^3\ ksi)(3\ in)(0.03\ in)}{2(18\ in)^2} = 12.083\ ksi$

$P_{max} = \dfrac{y_{max} \cdot 3EI}{L^3} = \dfrac{(0.03\ in)\,3(29\times10^3\ ksi)(1.83\ in^4)}{(18\ in)^3}$

$P_{max} = 0.8279\ kip$

$P_{max,FS} = P_{max} \cdot 1.5 = 1.242 \ kip$

$M = \frac{(1.242 \ kip)}{2}(18 \ in) = 11.177 \ kip \cdot in = R \cdot (2 \ in) \Rightarrow R = 5.589 \ kip$

$F = \sqrt{\left(\frac{1.242 \ kip}{2}\right)^2 + (5.589 \ kip)^2} = 5.623 \ kip$

$\Rightarrow D = 3/8 \ in \qquad t = 0.16 \ in$

$h = 4 \ in \Rightarrow e = \frac{h - d_v}{2} = 1 \ in \Rightarrow \frac{e}{D} = \frac{1 \ in}{3/8 \ in} = 2.67$

$R_{FBR} = 2.94 \Rightarrow t_{eff} = \frac{t}{R_{FBR}} = 0.054 \ in \Rightarrow \underline{\underline{t_{eff} = 0.063 \ in}}$

$F_{actual} = \frac{F}{1.5} = 3.749 \ kip$

$FS_{rivet} = \left(\frac{3.749 \ kip}{6.085 \ kip}\right)^{-1} = 1.62$ //

$FS_t = \frac{6.176 \ kip}{3.749 \ kip} = 1.65$ //

Safe //

//

B

Appendix B

B.1 MATLAB Code

```
% AerE 426, Lab Assignment 6 - Joint Design: Problem 1, Lucas Tavares

clear, clc, close all

% Define bolts positions (from gusset plate's bottom left corner)
bolts_coords = [[1, 5]; [3, 5]; [5, 5]; [7, 5]; [2, 3]; [4, 3]; [6, 3]; [1,
    1]; [3, 1]; [5, 1]; [7, 1]]; %in

% Calculate mean X and Y (since D is constant among all bolts)
X_bar = sum(bolts_coords(:,1)) ./ length(bolts_coords);
Y_bar = sum(bolts_coords(:,2)) ./ length(bolts_coords);


% Diameter assumed arbitrarily
D = 3/8; %in

% Calculate moment of inertia
I = sum(D * bolts_coords(:,1).^2) + sum(D * bolts_coords(:,2).^2) - X_bar *
    sum(D * bolts_coords(:,1)) - Y_bar * sum(D * bolts_coords(:,2));

% Applied loads with FS = 5
P_x = 5 * 1750; %lbs
P_y = 5 * 1250; %lbs

% Moment at centroid (X_bar, Y_bar) (positive counterclockwise)
M = P_x * Y_bar + P_y * (20 - X_bar); %lbs*in

% Calculate load at each rivet
P = D .* sqrt(((P_y / (D * length(bolts_coords))) + (M .* (bolts_coords(:,1)
     - X_bar) ./ I)).^2 + ((P_x / (D * length(bolts_coords))) + (M .* (
    bolts_coords(:,2) - Y_bar) ./ I)).^2); %lbs

% Retrieve maximum load at rivets
P_max = max(P); %lbs
fprintf('\nMax load at joints: %.4f lbs\n\n', P_max)
```

```

% From table, chose rivet with 5/16 in and A-286 material; F_rivet = 7375
% lbs
D = 5/16; %in
F_rivet = 7375; %lbs

% Max load at joints without FS of 5
P_max_actual = P_max / 5; %lbs
fprintf('Actual max load at joints: %.4f lbs\n\n', P_max_actual)

% Calculate the FS for the rivet
FS_rivet = F_rivet / P_max_actual;
fprintf('FS_rivet: %.4f \n\n', FS_rivet)

% Minimum distance between rivet hole and wall e
e = 1; %in

e_over_D = e / D;
fprintf('e/D: %.4f \n\n', e_over_D)

% From table, F_bearing_yield = 346 ksi
F_bearing_yield = 346000; %psi

% Calculate allowable P for bearing
t = 3 / 8; %in
P_allowable_bearing = F_bearing_yield * D * t; %lbs

% Calculate FS for bearing
FS_bearing = P_allowable_bearing / P_max_actual;

fprintf('FS_bearing: %.4f \n\n', FS_bearing)
```